Cloud Computing and Big Data

Big Data Application: Amazon Web Services

- Millions of customers taking advantage of cloud computing:
  - Lower costs
  - More agile applications
  - Faster innovation
- Provide storage capacities, Hadoop, Pig, Hive, Spark
- Allow business organizations to perform:
  - Interactive and iterative processing
  - Real-time analytics
  - Operational intelligence modeling
  - Dashboards
  - Data visualizations
  - Customers: J.D. Power, Netflix, Chick-fil-A, 3M, Georgia-Pacific, Pinterest, T-Mobile, Epic Games, Equinox, Adobe, and Pfizer

Chapter Objectives:

- Explain the concept of cloud computing and its primary services to organizations
- Discuss the advantages of using cloud computing services for Big Data Analytics
- Set up a Hadoop cluster using the EMR service of AWS

Introduction

- Big Data and cloud computing technologies provide:
  - Fast and elastic cluster deployment
  - Efficient data processing
  - Lower cost
  - Simpler solutions

What Is Cloud Computing?

- Because of **increasing** *volume, velocity, variety* of data:
  - **Cost and feasibility** of maintaining physical data centers => **burden** for business organizations
  - Organizations are switching from traditional on-premise computing services to cloud computing storage and services
- Organizations can access technology services:
  - Computing power
  - Storage
  - Databases
  - As-needed basis from anywhere in the world

What Is Cloud Computing? (cont.)

- Healthcare centers use the cloud to:
  - Store patient information, analyze it
    - To offer more personalized treatments to their patients
- Financial services companies:
  - Utilize powerful cloud-based algorithms
    - To detect and prevent fraud in real time
- Logistics and supply chain organizations:
  - Use network-based cloud services
    - To track the delivery of their products in real time and monitor location accuracy

Cloud-Based Applications Categories

- Software as a Service (SaaS):
  - Offers end-user applications:
    - Web-based email
    - Online-training
    - Sales management
    - Customer relationship management
    - Financial management
    - Human resource management
    - Billing
    - Collaboration software
  - SaaS providers include: Salesforce, Oracle, SAP, Intuit, and Microsoft

Cloud-Based Applications Categories (cont.)

- Platform as a Service (PaaS):
  - Offers hardware and software
  - Does not replace the entire IT infrastructure
  - Allows organizations to focus on the deployment and management of applications

Cloud-Based Applications Categories (cont.)

- Infrastructure as a Service (IaaS):
  - Offers full support (hardware and software) along with the maintenance-related tasks
  - Provides the highest level of flexibility and control over the organization's IT resources

Cloud Computing Services

- Data backup
- Disaster recovery
- Email
- Storage
- Networking software development and testing
- Big Data Analytics
- Databases
- Providers offer:
  - Free access to their data platforms on a trial basis—12 months at the time this book was written
  - Academic partnerships that allow instructors and students to receive free access credits during a given academic course

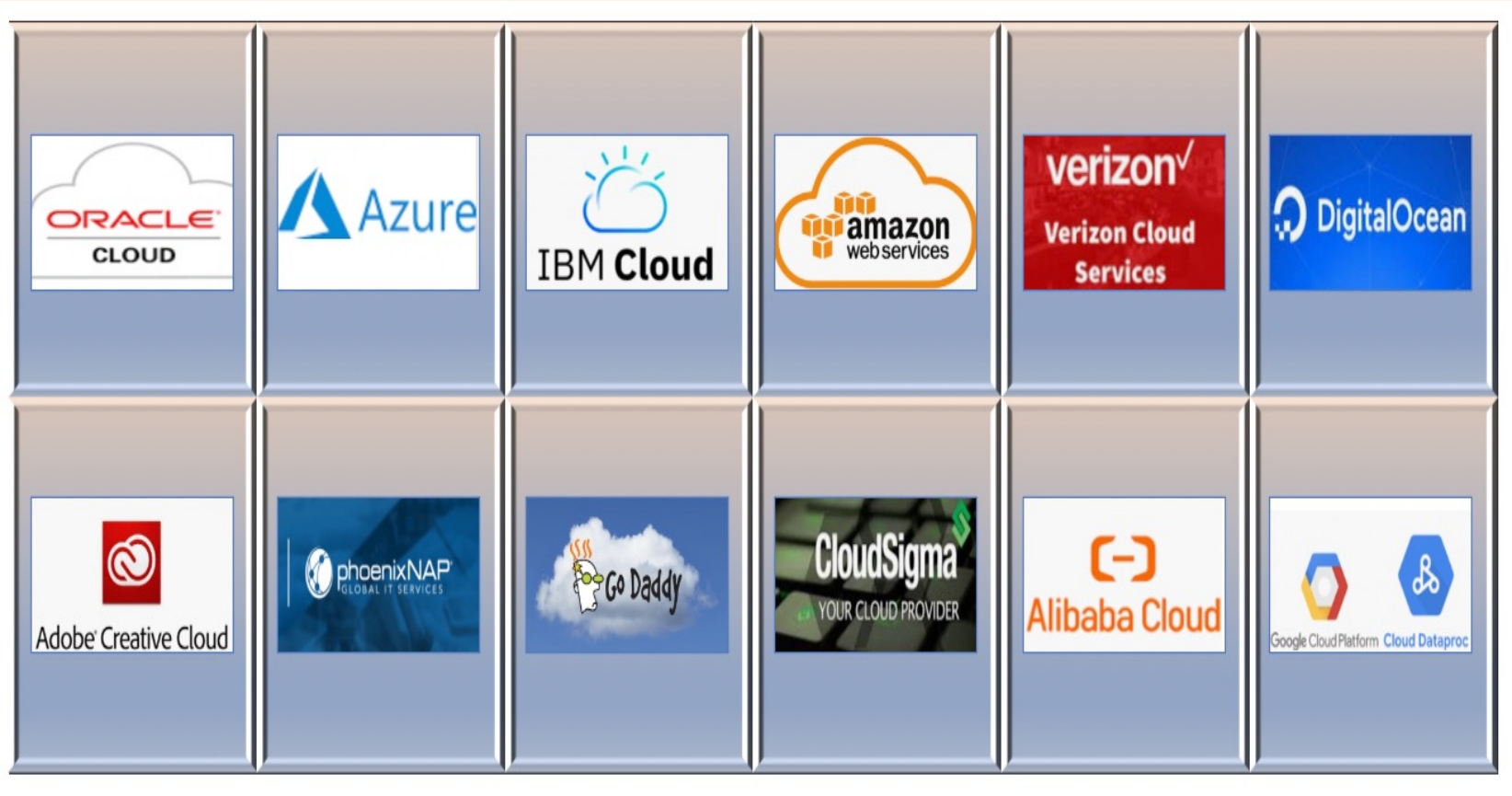


Figure 2-1: Major Cloud Computing Providers

Microsoft Azure

- Offers Big Data Analytics platform to perform large-scale workloads that require hundreds or thousands of cores
- Using Microsoft Azure, Big Compute architecture:
  - Configures a pool of virtual machines
  - Uploads the applications and data files
- The batch service:
  - Provisions the virtual machines
  - Assigns and runs tasks
  - Monitors the progress
  - Automatically scales out the number of virtual machines in response to the workload
- Big Data solutions from Azure perform:
  - Batch processing, real-time processing, interactive exploration, predictive analytics, and machine learning

Google Cloud Dataproc

- Offers several features for Big Data processing and storage
- Allows for cluster autoscaling in a short amount of time
- The cluster can run open-source Big Data processing software:
  - Apache Hadoop
  - Apache Spark
- Clusters offer a high degree of security and encryption

Amazon Web Services (AWS)

- Offers many cloud-based services
- Elastic MapReduce (EMR) component:
  - Provides fast, easy, and cost-effective way to run:
    - Apache Hadoop
    - Apache Spark
  - Supports other Hadoop tools:
    - Hive, Pig, Sqoop, Kinesis, HBase
- Offers scalable and cost-effective "data lakes" for "loading, transforming, securing, and auditing access to" Big Data

Advantages of Using Cloud Computing for Big Data Analytics

1. Agility
2. Elasticity
3. Cost Savings
4. Big Data Security

1-Agility

- Cloud-based services give organizations access to a broad range of Big Data technologies
- Organizations can scale-out infrastructure services:
  - Storage, compute
  - Databases
  - Data lakes, analytics
  - Machine learning
- Organizations can deploy the latest version of Big Data software in a short amount of time

2-Elasticity

- Using cloud computing services, organizations:
  - Acquire the number of resources that they need *when they need them*
  - Can scale-out or scale-in to either *grow* or *reduce* their Big Data Analytics capacity as business needs change

3-Cost Savings

- Organization pays for Big Data technologies as it consumes them, just as it would pay for other utilities
- The *variable expenses* are **lower** than what organizations would pay for in-house Big Data Analytics because of the **economies of scale**

4-Big Data Security

- Traditionally, companies would have to:
  - Set up their user access policies
  - Install firewalls and antivirus software to guard them against cyber threats
- Cloud Computing services:
  - Offer a broad set of policies, technologies, and controls to protect data, applications, and infrastructure
  - Run on a worldwide network of secure data centers
  - Regularly upgrade with the latest generation of computing hardware

Challenges of Cloud Computing for Big Data Analytics

1. Customer and employee privacy
2. Information assurance and compliance
3. Data availability

1-Privacy

- Organizations must always ensure that their data and information are protected
- Both traditional and Big Data sets often contain sensitive information:
  - Addresses
  - Credit card details
  - Social Security numbers
- Many organizations might feel a lack of control and some discomfort when migrating these datasets to the cloud
- However, evaluating security protocols with the cloud service provider might mitigate these concerns

2-Compliance

- Many organizations might:
  - Feel that if they have their data in the cloud, they will have less control of compliance with regulations:
    - HIPAA
    - PCI
  - Have concerns about compliance issues when their Big Data is stored and processed:
    - Outside the domain of their own information technology department
    - Outside the organizational boundaries

3-Data Availability

- High dependence on the Internet means that the system is prone to:
  - Service interruptions
  - Latency issues
- Although migrating Big Data to the cloud presents some hurdles, the opportunities outweigh the challenges

Tutorial A:

Setting Up a Hadoop Cluster on AWS

Tutorial A: Setting Up a Hadoop Cluster on AWS

- In this tutorial, you will:
  - Register for an AWS account
  - Set up an Elastic MapReduce (EMR) cluster
  - Connect to the master node using the Secure Shell (SSH) protocol
  - Set up a File Transfer Protocol (FTP) program to move files to the cluster

Tutorial A: Registering for an AWS account

- Simply google "AWS free tier" or visit https://aws.amazon.com/free/
- Click the "Create a Free Account" button
- If you are an instructor:
  - Visit https://aws.amazon.com/education/awseducate/ to sign up for an educational account
  - Once your account is approved:
    - Create a classroom
    - Provide your students with free and easy access to the AWS Console for homework, labs, and projects

Tutorial A: Registering for an AWS account (cont.)

- If you are a student:
  - Receive an email from AWS services with a registration link
    - Complete the personal information
    - Receive another email confirmation that your application is approved
    - Sign in to the AWS Educate site by following this link: https://aws.amazon.com/education/awseducate/

Tutorial A: AWS Management Console

- Click "My Classrooms" tab
  - Click "Go to the classroom" to access your AWS Educate Account
    - Start and access the variety of AWS services
    - "AWS Console" button => AWS Management Console
      - Allows you to access many services provided by Amazon Web Services

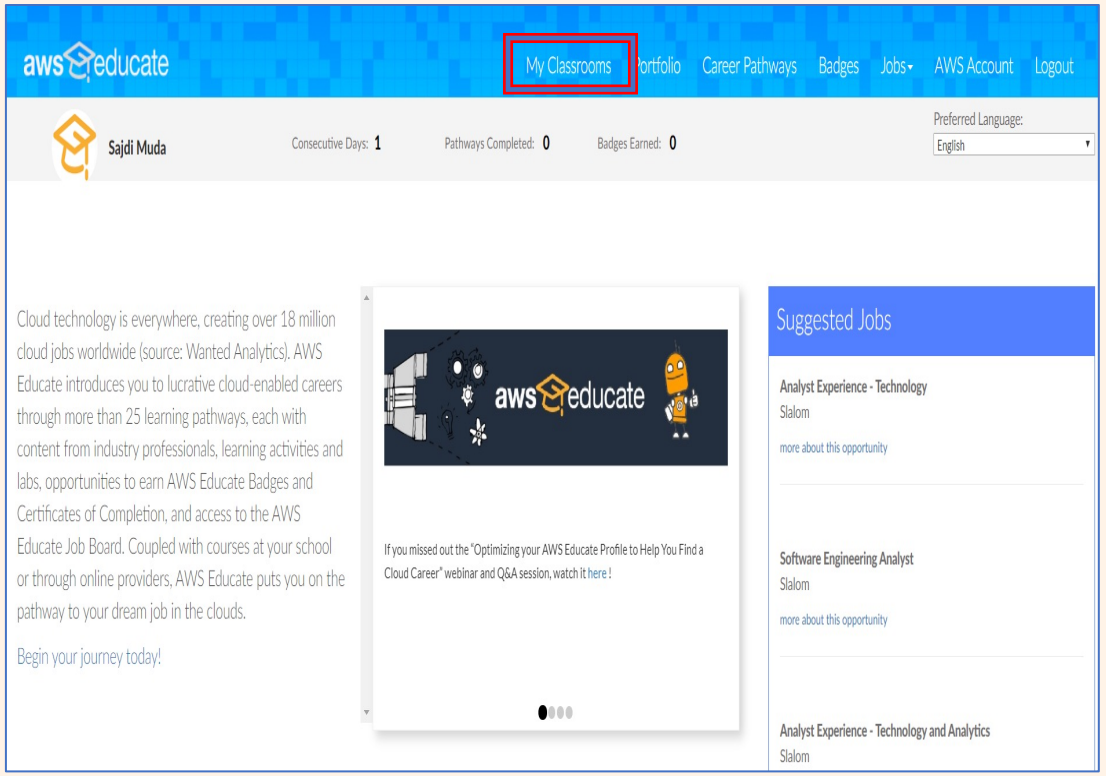


Figure 2-2: AWS Educate Student Profile

Go to AWS Console

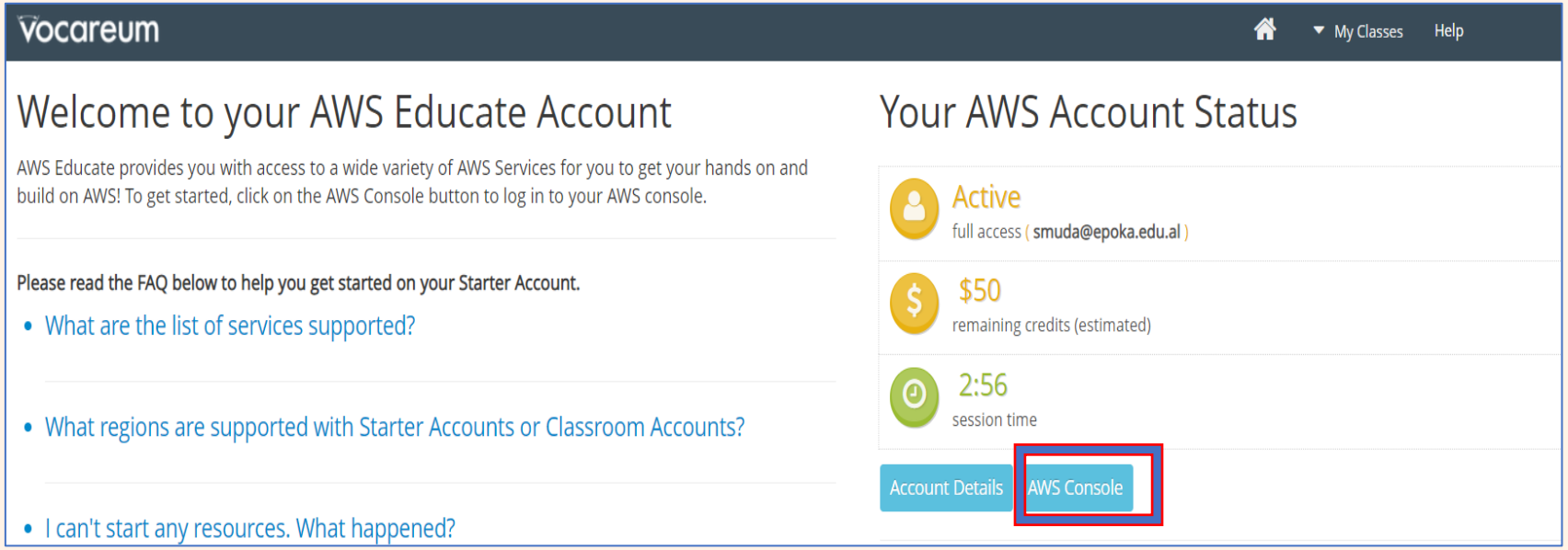


Figure 2-3: AWS Educate Account

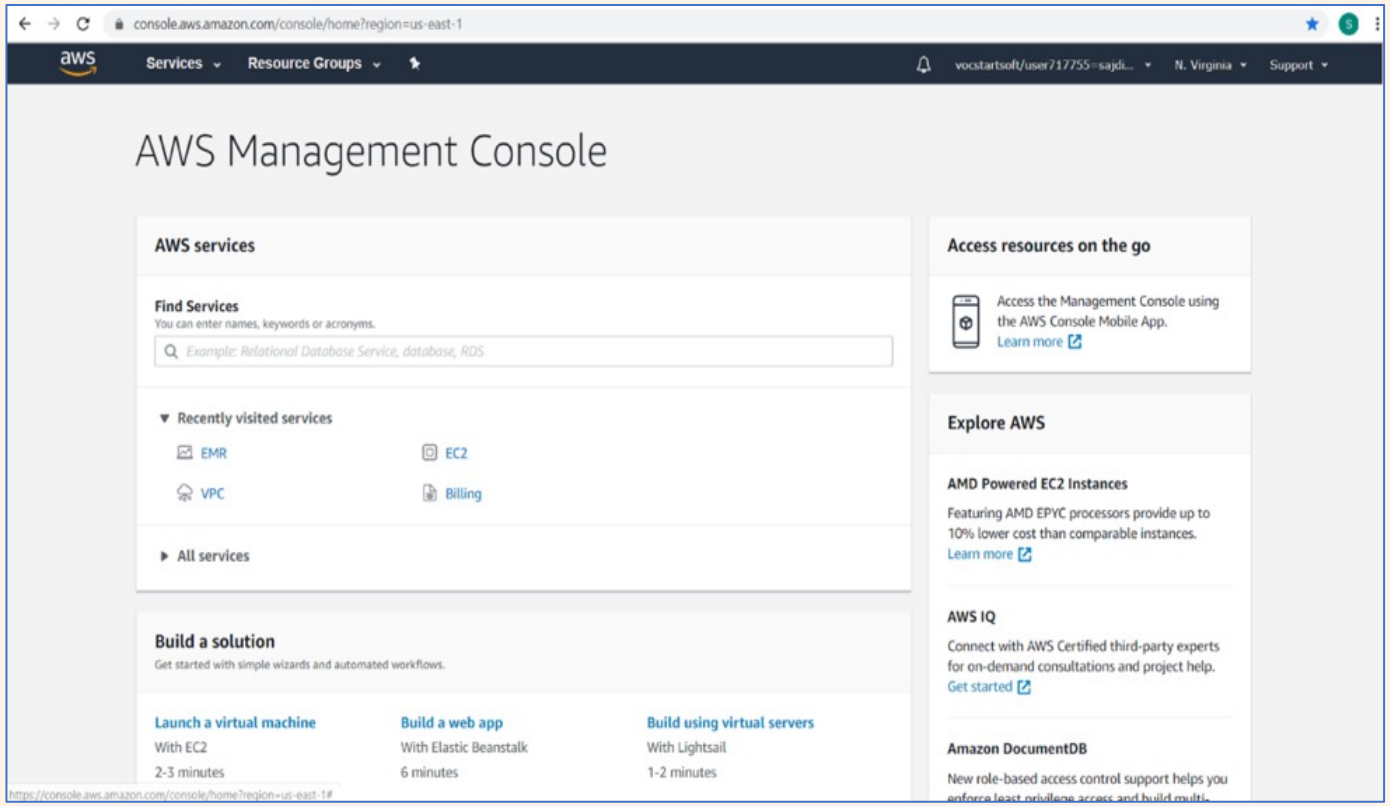


Figure 2-4: The AWS Management Console

Step 1: Create a Security Key Pair

- From "Services" tab:
  - Under "Compute"
    - Click "EC2"

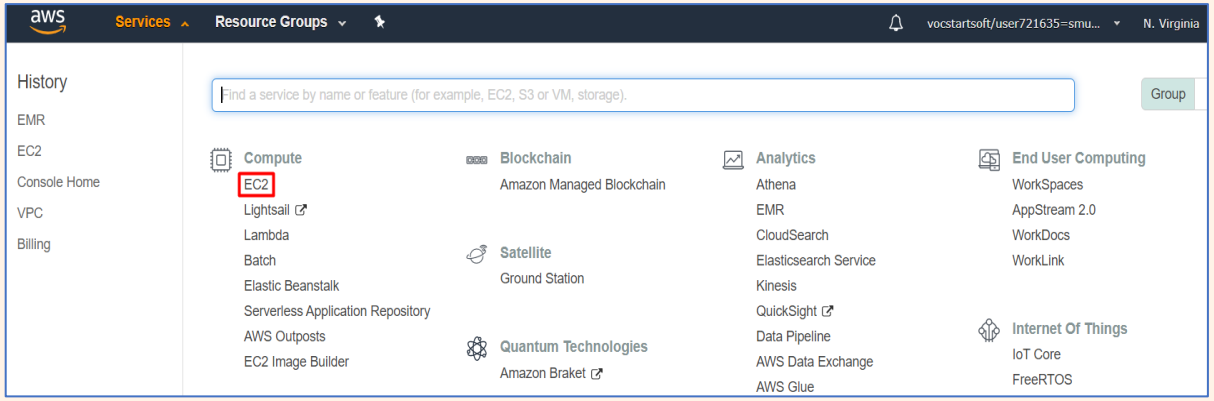


Figure 2-5: Selecting Elastic Cloud Computing (EC2) Services

Step 1: Create a Security Key Pair (cont.)

- In the navigation pane
  - Click "Key Pairs"
    - Click "Create key pair"

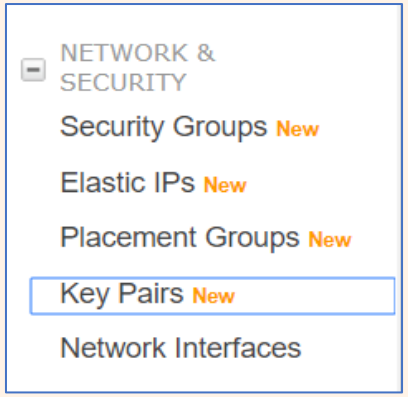


Figure 2-6: Selecting "Key Pairs" from the Menu

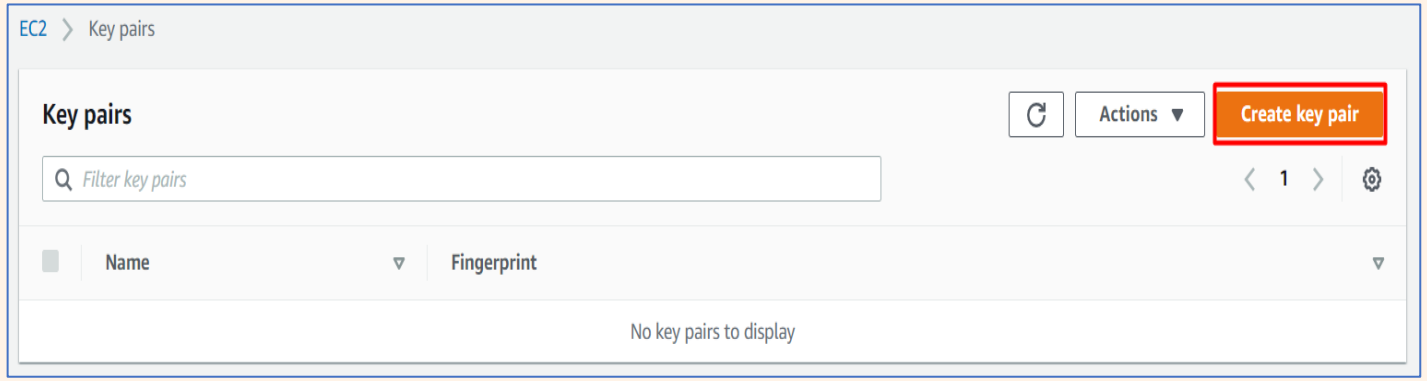


Figure 2-7: Selecting "Create key pair"

Step 1: Create a Security Key Pair (cont.)

- Name the key:
  - Windows OS requires you to save the key with a "ppk" file format
  - macOS requires you to save with a "pem" extension
- Key is downloaded to a folder on your computer you specify once you click "Create key pair"

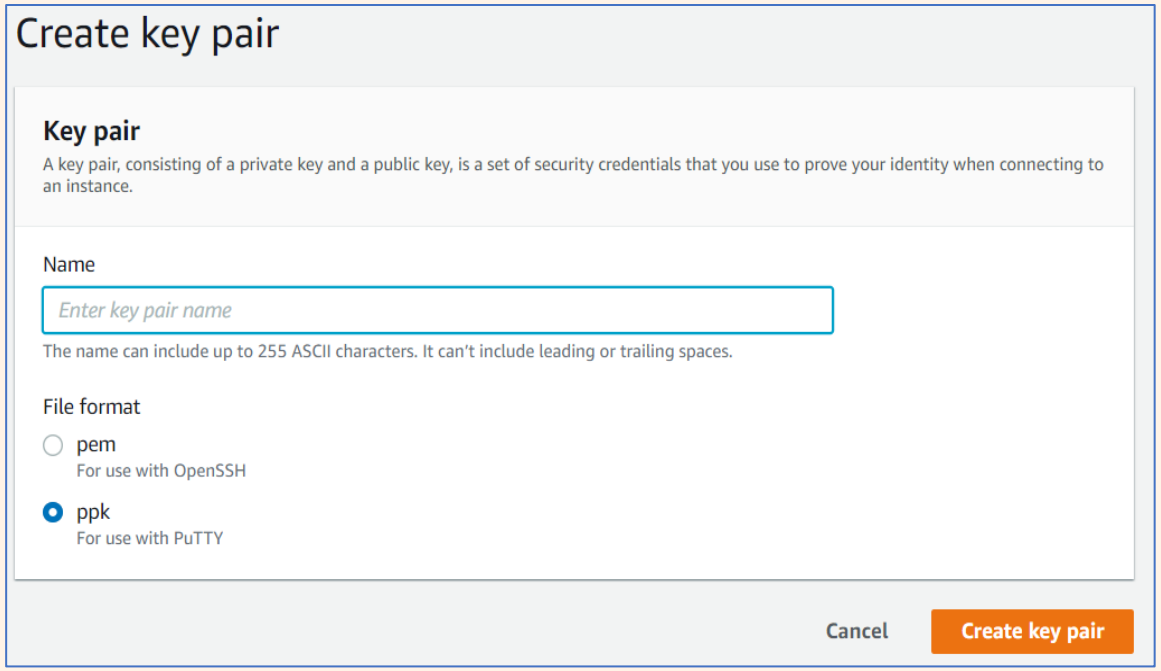


Figure 2-8: Naming and Creating the "Key pair" File

Step 2: Setting up the EMR Cluster

- From "Services" tab
  - Click "EMR" under "Analytics”

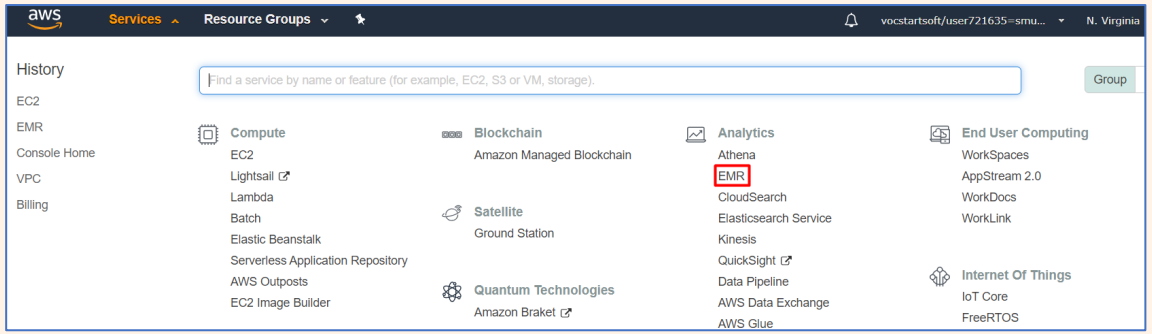


Figure 2-9: Selecting the EMR Services

Step 2: Setting up the EMR Cluster (cont.)

- Select “Create cluster”
- Click “Go to advanced options”
- Check all the boxes next to the software packages you want in the cluster

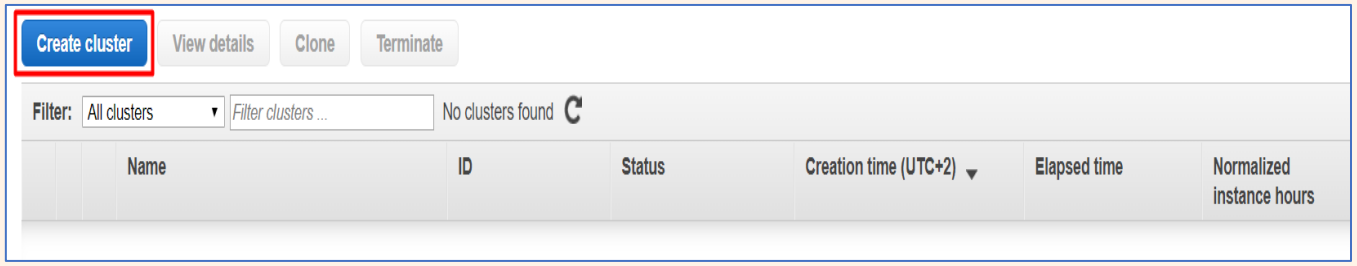


Figure 2-10: Selecting "Create cluster"

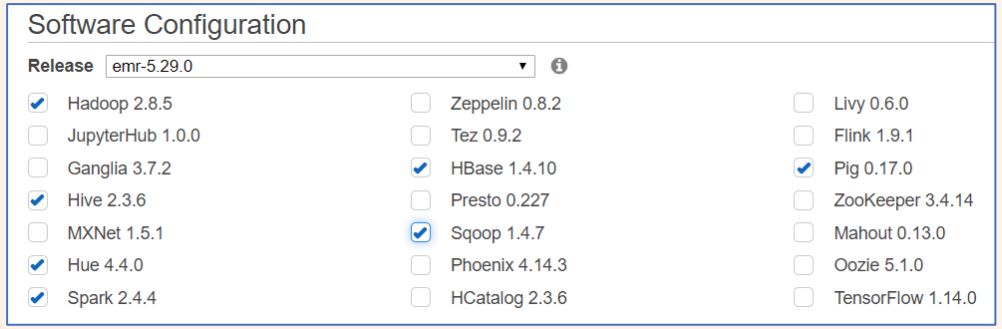


Figure 2-11: Selecting Hadoop and its Ecosystems in the EMR Cluster in advanced options

Step 2: Setting up the EMR Cluster (cont.)

- Change hardware configurations
- OR
- Leave the default parameters

Click “Next”

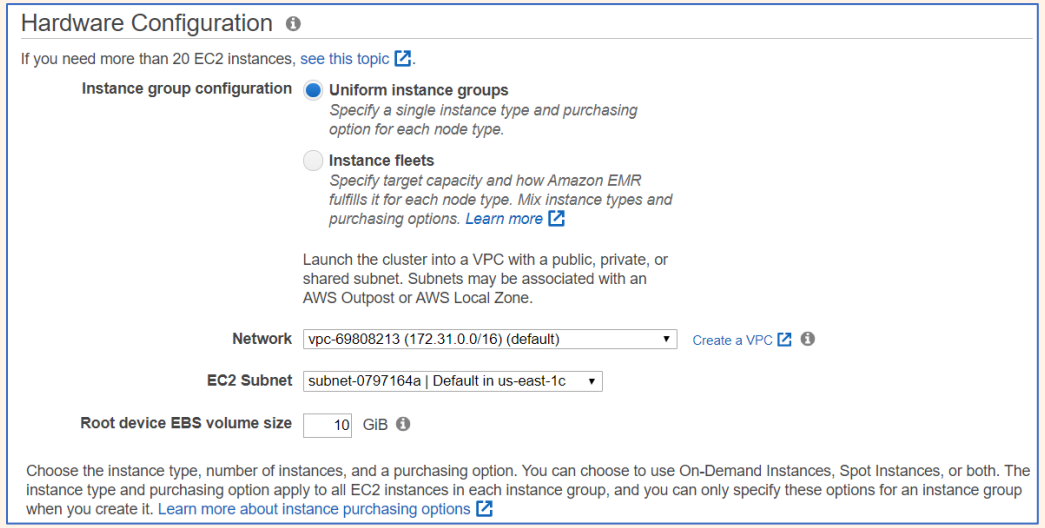


Figure 2-12: Selecting Hardware Configuration Options

Step 2: Setting up the EMR Cluster (cont.)

- Change general options
- OR
- Leave the default parameters

Click “Next”

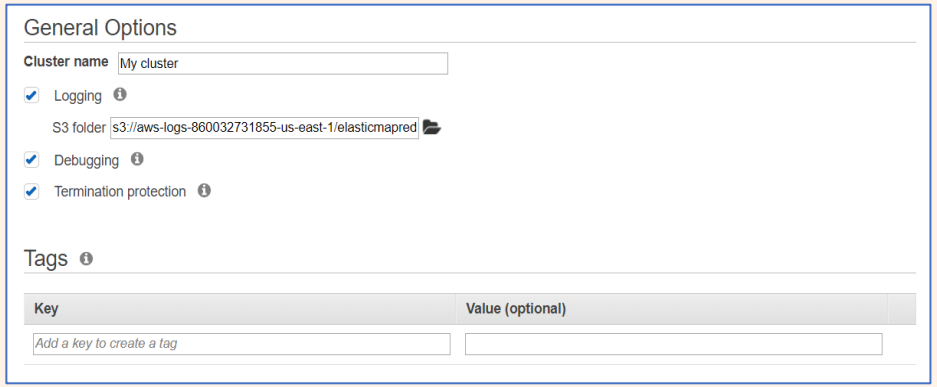


Figure 2-13: Selecting General Options

Step 2: Setting up the EMR Cluster (cont.)

- Security Options – EC2 key pair:
  - Select the key file that you created in Step 1
- Click "Create cluster"

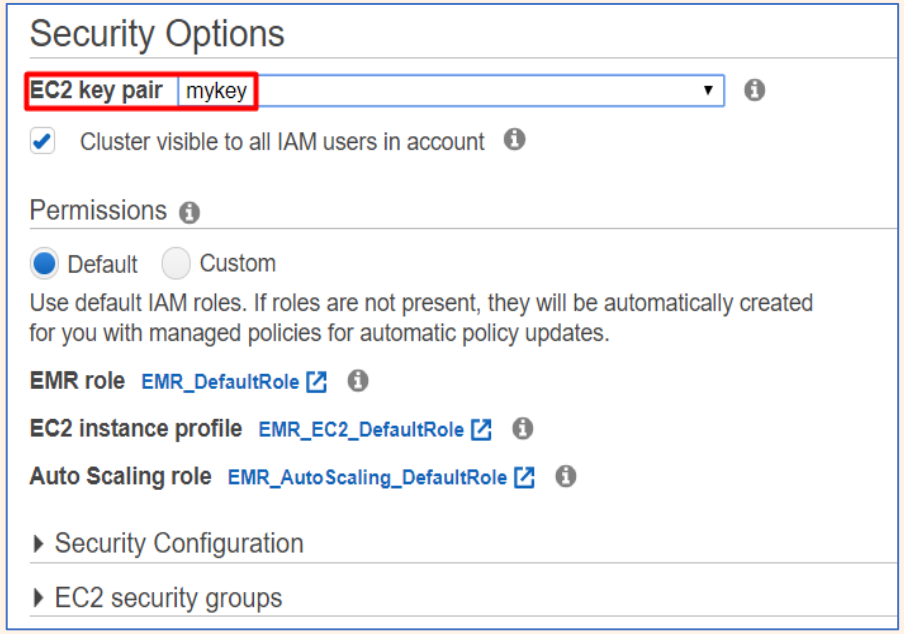


Figure 2-14: Selecting the Key-Pair File in Security Options

Step 2: Setting up the EMR Cluster (cont.)

- Wait for the status to change from ‘Starting’ to ‘Running’ to ‘Waiting’

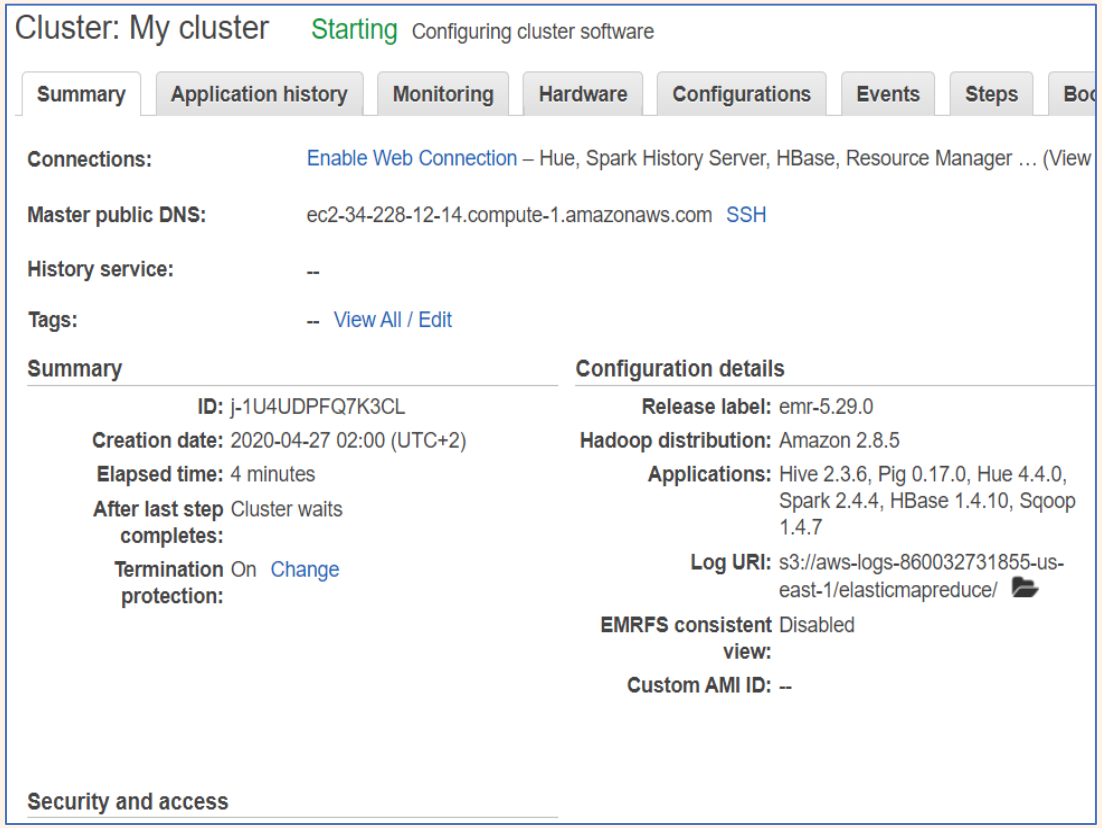


Figure 2-15: Starting the Cluster

Step 3: Allowing SSH Traffic into the Cluster

- Under "Security and access"
  - Click the "Security groups for Master"

Security and access

Key name: mykey
EC2 instance profile: EMR_EC2_DefaultRole
EMR role: EMR_DefaultRole
Auto Scaling role: EMR_AutoScaling_DefaultRole
Visible to all users: All Change
Security groups for Master: sg-01cc3b88cf473ed57 (ElasticMapReduce-master)
Security groups for Core & Task: sg-00b021f1e39a46248 (ElasticMapReduce-slave)

Figure 2-16: Selecting the Security Groups for Master

Step 3: Allowing SSH Traffic into the Cluster (cont.)

- Select ”Inbound rules”

- Click ”Edit inbound rules”

Figure 2-17: Editing Inbound Rules

Step 3: Allowing SSH Traffic into the Cluster (cont.)

- Make sure that the "SSH" type is listed
- If not:
  - Go to the bottom of the page
  - Click "Add rule"
  - Select:
    - "SSH" for the type
    - 22 as the port
    - 0.0.0.0/0 as the "IP" address
  - Click "Save rules"
- This rule allows SSH (TCP port 22) traffic from a source that includes the client where the SSH connection originates

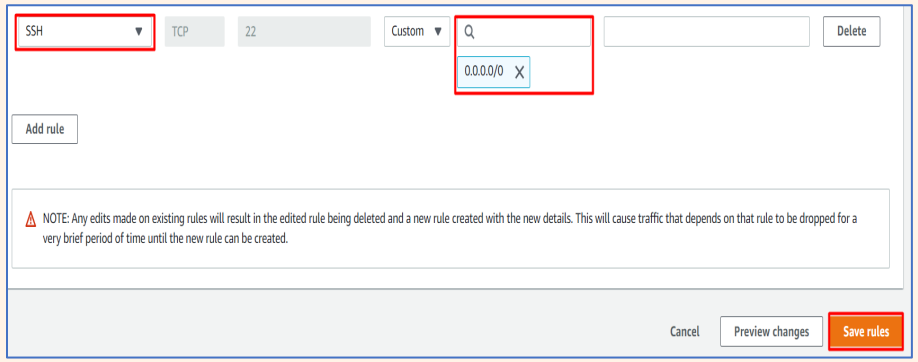


Figure 2-18: Adding SSH Rule

Tutorial A: Using SSH to Connect to the Cluster

- SSH:
  - Network protocol
  - Used to create a secure connection to a remote computer
  - After making a connection, the terminal on the local computer behaves as if it is running on the remote computer
  - Issued commands run locally on the remote computer
  - The command output from the remote computer appears in the terminal window
  - When using SSH with AWS, the connection is made to an EC2 instance:
    - Virtual server running in the cloud

Tutorial A: Using SSH to Connect to the Cluster (cont.)

- Click “Connect to the Master Node Using SSH”

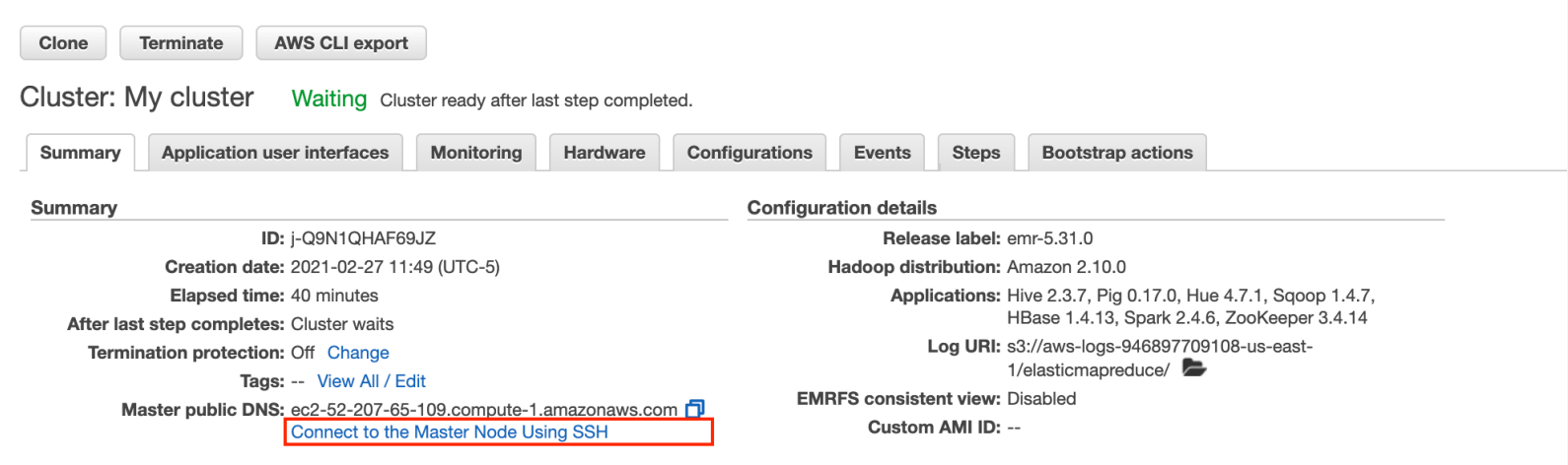
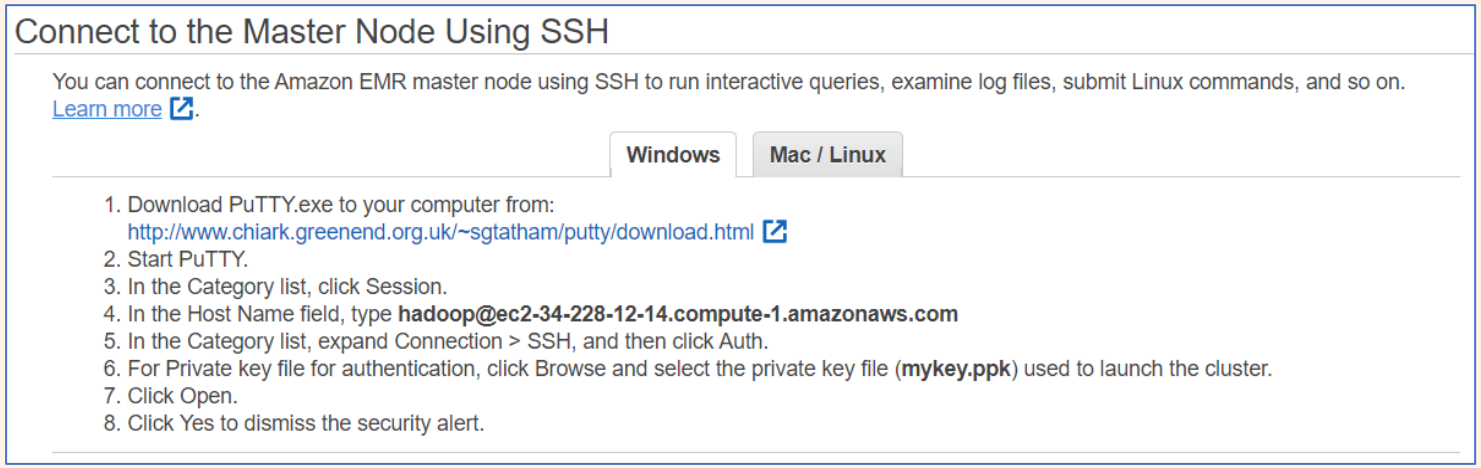


Figure 2-19: Selecting the Public DNS for the Master Node

Tutorial A: Using SSH to Connect to the Cluster (cont.)

- Follow the steps shown in the window for either the Windows or Mac operating system



Connect to the Master Node Using SSH

You can connect to the Amazon EMR master node using SSH to run interactive queries, examine log files, submit Linux commands, and so on. Learn more.

Windows | Mac / Linux

1. Download PuTTY.exe to your computer from:
   http://www.chiark.greenend.org.uk/~sgtatham/putty/download.html
2. Start PuTTY.
3. In the Category list, click Session.
4. In the Host Name field, type **hadoop@ec2-34-228-12-14.compute-1.amazonaws.com**
5. In the Category list, expand Connection > SSH, and then click Auth.
6. For Private key file for authentication, click Browse and select the private key file (**mykey.ppk**) used to launch the cluster.
7. Click Open.
8. Click Yes to dismiss the security alert.

Figure 2-20: Details for Connecting to the Master Node via Windows

Connecting via macOS, Linux, Unix

- Open a terminal window
  - Choose Applications > Utilities > Terminal in macOS
  - On other Linux distributions, choose Applications > Accessories > Terminal
- Set the permissions so that the key owner has access to the file:
  - $ chmod 400 /<path/to/key file>
- Connect to the master node using the terminal:
  - $ ssh -i ~/</path/key_file.pem> <hadoop@public master DNS>

Connecting via Windows

- Download PuTTY.exe from http://www.chiark.greenend.org.uk/~sgtatham/putty/download.html
- Start PuTTY by:
  - "Category" => "Session" => "Host Name" => type the master public DNS

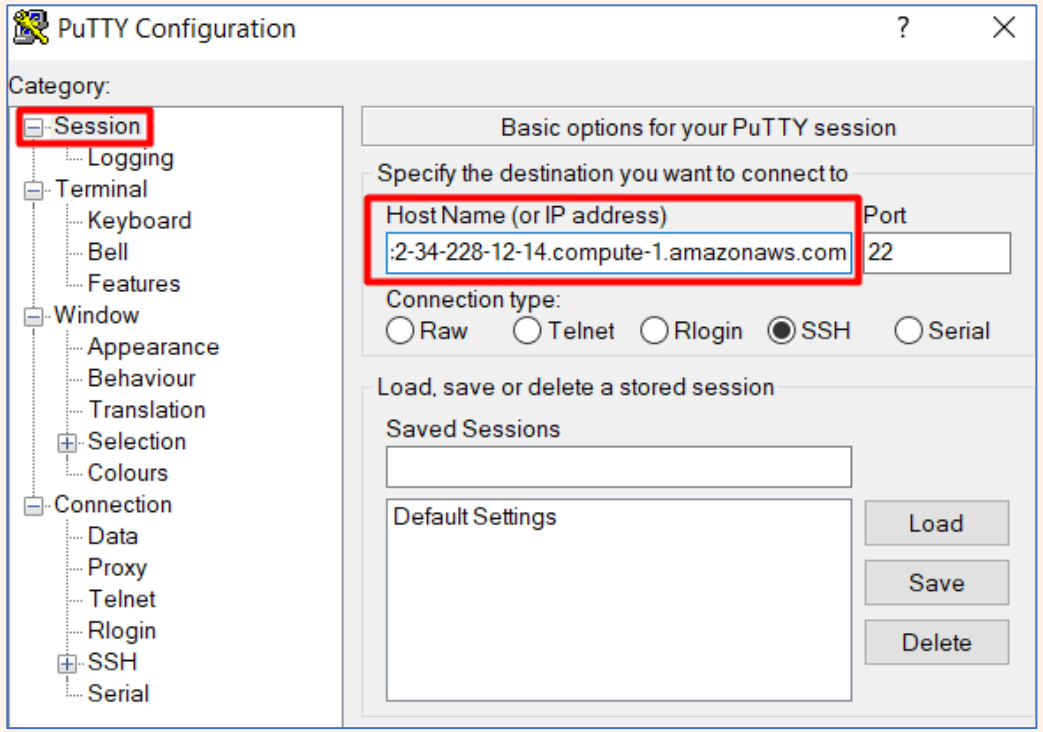


Figure 2-21: Starting a Session in PuTTY

Connecting via Windows (cont.)

- Set the private key by:
  - "Category" => expand "Connection > SSH" => click "Auth"
    - "Private key file for authentication" => click "Browse" => select the private key file you created earlier and downloaded to your computer
    - Launch the cluster by clicking “Open”

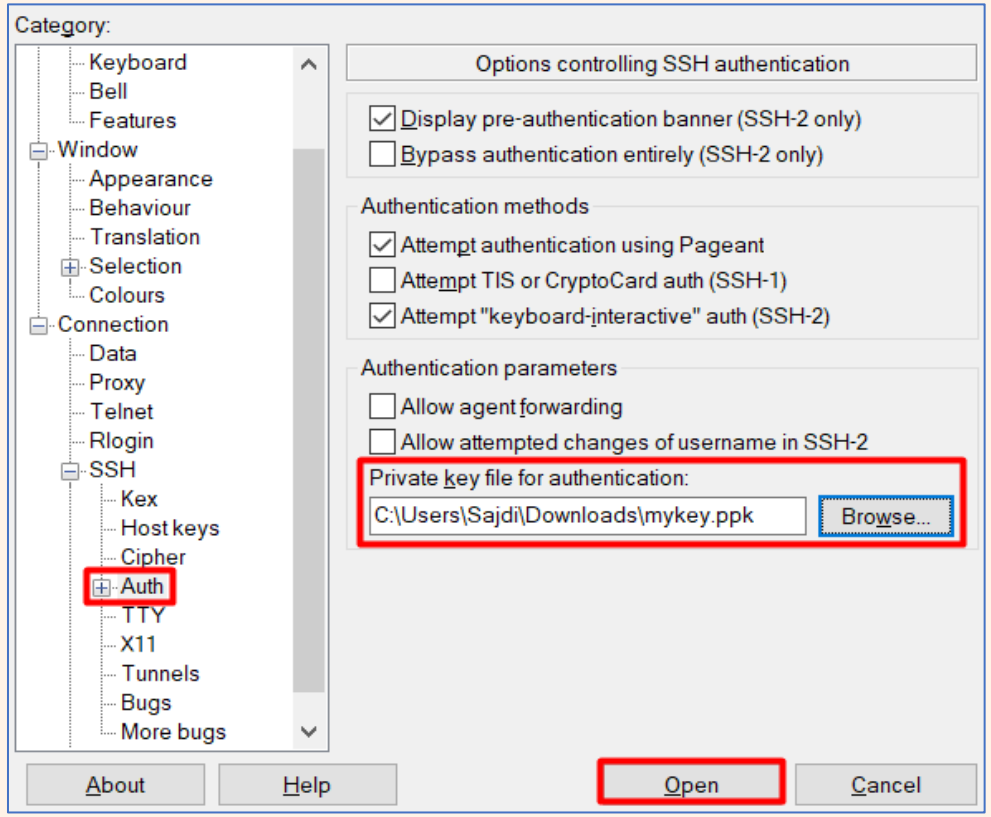


Figure 2-22: Setting the Private Key

Connecting via Windows (cont.)

- PuTTY starts, and the EMR cluster is ready to interact

```
Using username "hadoop".
Authenticating with public key "imported-openssh-key"
Last login: Sun Apr 26 17:45:29 2020

       __|  __|_  )
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      ___|\___|___|

https://aws.amazon.com/amazon-linux-ami/2018.03-release-notes/
21 package(s) needed for security, out of 40 available
Run "sudo yum update" to apply all updates.

EEEEEEEEEEEEEEEEEEEE MMMMMMMM           MMMMMMMM RRRRRRRRRRRRRRR
E::::::::::::::::::E M:::::::M         M:::::::M R::::::::::::::R
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EEEEEEEEEEEEEEEEEEEE MMMMMMM              MMMMMMM RRRRRRR      RRRRRR

[hadoop@ip-172-31-68-16 ~]$
```

Figure 2-23: Starting the EMR Cluster

Tutorial A: Setting Up the File Transfer Protocol

- User-friendly tools allow us to transfer data to a host
- FileZilla supports Windows, Linux, and Mac systems
- WinSCP: open-source file transfer client for Windows
- The tools support:
  - File Transfer Protocol (FTP)
  - Secure File Transfer Protocol (SFTP)
  - Other file transfer protocols

Tutorial A: Setting Up the File Transfer Protocol (cont.)

- Open WinSCP:
  - Create a new site that you can use to connect with the host
  - Choose the SFTP file protocol, which runs over the SSH protocol and supports the security and authentication functionality of SSH
  - For the host name, use the master public DNS of the created cluster, with port number 22, which is the port of SFTP, and provide the username; the username is "hadoop"

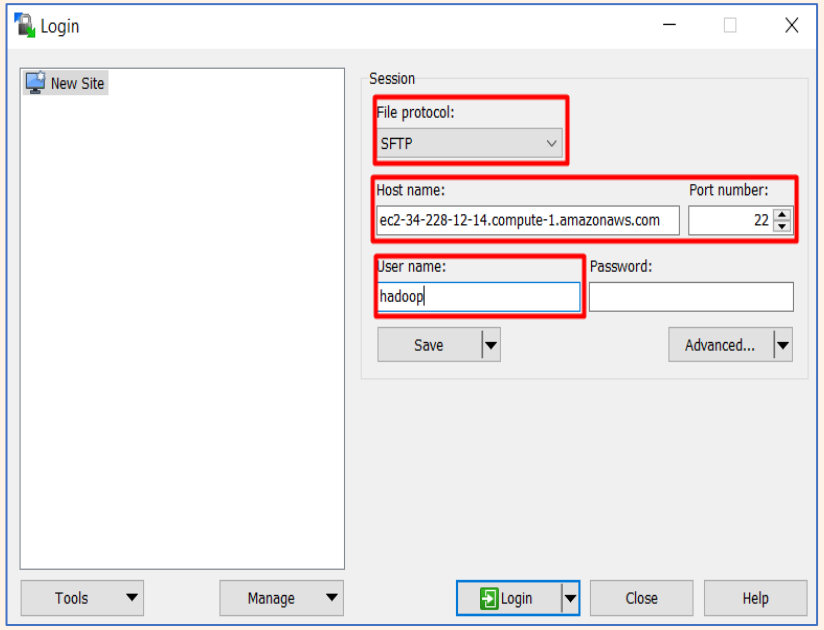

Figure 2-24: New Site Configuration Options in WinSCP

Tutorial A: Setting Up the File Transfer Protocol (cont.)

- Click "Advanced..."
  - "SSH > Authentication" tab => choose the .ppk key file from the computer under "Private key file"
  - The actual path to the .ppk key file should match the directory path where you have located your key file
  - Click "OK" => the key connects the local machine

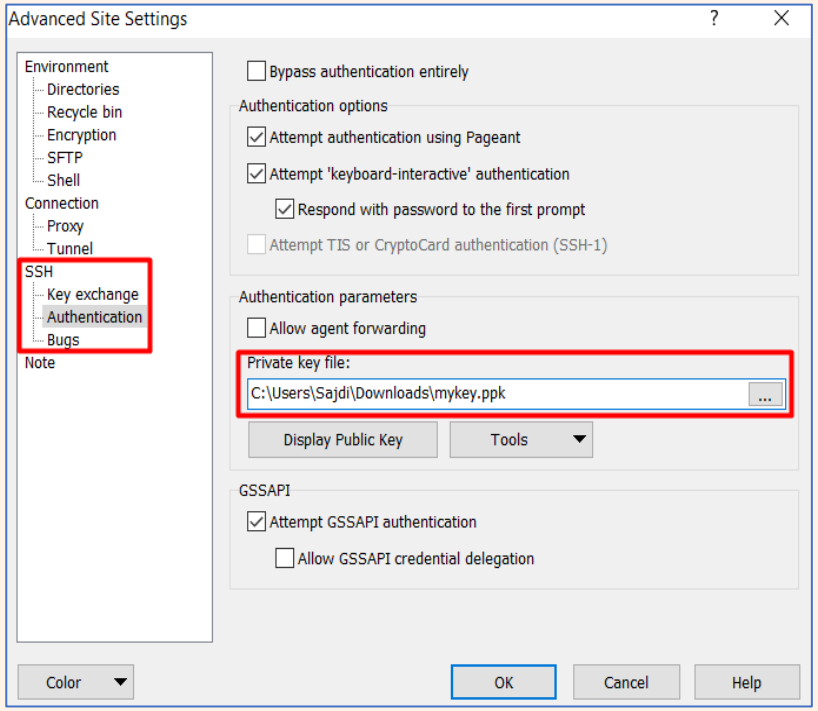

Figure 2-25: Setting Up the Private Key in WinSCP

Tutorial A: Setting Up the File Transfer Protocol (cont.)

- Click "OK"
- Click "Yes" to the warning
- WinSCP is connected to the EMR host
- Perform the process of file transfer between the device and the Hadoop cluster
- WinSCP lists:
  - The device files in the left pane
  - The files in the Hadoop cluster home directory in the right pane

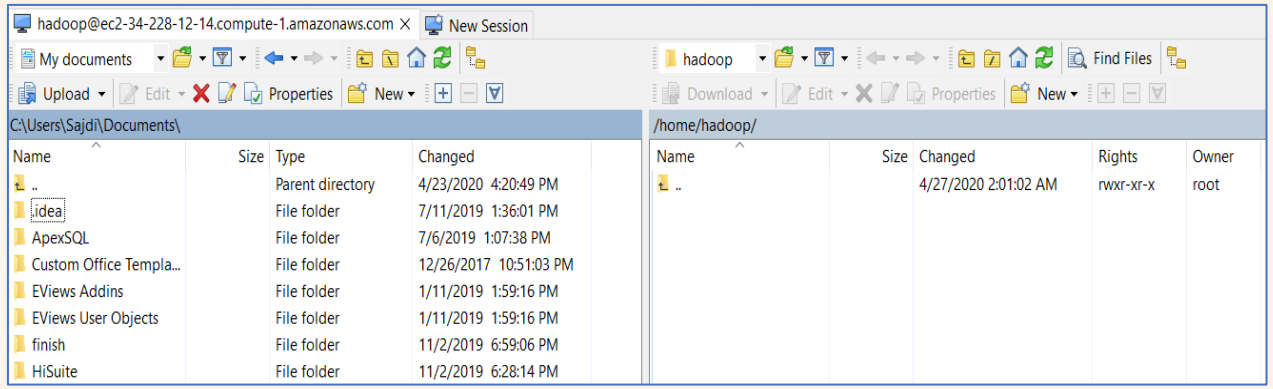

Figure 2-26: File Transfer in WinSCP

Summary and Takeaways

- Business organizations continue to transform their IT capabilities from traditional, in-house computing to cloud storage and computing
- Using cloud computing, organizations can store and analyze a high volume of data records from a variety of data sources at a high speed of data production
- Performing Big Data Analytics:
  - Allows an organization to expand its IT infrastructure and to deploy the latest Big Data technologies
  - Organization can grow or reduce its Big Data Analytics capacity as business needs change
  - The affordable scalability feature of the cloud-based models is the best computing option for Big Data Analytics
  - When using cloud computing technology, organizations can take advantage of a utility-based, pay-as-you-go model, which leads to significant cost savings
  - Organization can improve Big Data security

Summary and Takeaways (cont.)

- Consider the need for:
  - Better protection of customer and employee privacy
  - Establishing information assurance and compliance
  - Offering better data availability and consistency
- Big Data Analytics can utilize cloud-based services:
  - Software as a Service (SaaS)
  - Platform as a Service (PaaS)
  - Infrastructure as a Service (IaaS
- Organizations can rent access to software, platforms, or complete infrastructure support available on the cloud
- Services offered by many cloud-based providers:
  - Microsoft Azure, Google Cloud Dataproc, and Amazon Web Services, among others
  - Offer free access to their data platforms on a trial basis to professionals and free access credits to students and professors

Summary and Takeaways (cont.)

- Cloud computing offers several advantages for Big Data Analytics
- It provides more storage for a high volume of data and more computing capabilities for advanced processing and analysis
- Organizations can choose different levels of cloud computing services, such as end-user applications, hardware, software, or IT resources in general
- Small and medium-sized businesses can benefit significantly from cloud-based services because their own computing and storage resources are limited
- AWS offers free access to professionals, students, and educators:
  - As you register for these services using the tutorial in this chapter, you may select storage and computing resources as provided by the EMR component of AWS
  - To make the best use of the free credits, make sure to terminate the EMR cluster at the end of your exercise practices
  - At the start of another session, you can quickly start a new cluster by merely cloning an existing terminated cluster